Syllabus Year: 2023-2024

CHRIST (Deemed to be University), Bengaluru - 560029

End Semester Examinations - November 2025

Bachelor of Computer Applications SEMESTER - II

Course Code: BCA104-2 **Max Marks: 100**

Course Name: DISCRETE MATHEMATICS **Duration: 03:00**

General Instructions:

- All rough work should be done in the answer script. Do not write or scribble in the question paper except your register number.
- Verify the Course code / course title and number of pages of questions in the question paper.
- Make sure your mobile phone is switched off and placed at the designated place.
- Malpractices will be viewed very seriously.
- Answers should be written on both sides of the paper in the answer book. No sheets should be detached from the answer book.
- Answers without the question numbers clearly indicated will not be valued. No page should be left blank in the middle of the answer book.

Course Outcome: Students will be able to:

CO1: Demonstrate a working knowledge of set notation and elementary set theory and recognize the connection between set operations and logic.

CO2: Apply the different properties of injections, surjections, bijections, compositions, and inverse functions.

CO3: Understand different types of matrices and operations among matrices.

CO4: Determine when a relation is reflexive, symmetric, antisymmetric, transitive, apply the properties of equivalence relations and partial orderings, and explain the connection between equivalence relations.

CO5: Effective express the elementary concepts and results of Number theory.

Section A **3*8=24**

1)	Examine and differentiate between a reflexive relation and an antisymmetric relation using two example relations defined on the set {1, 2, 3}. Also, determine if either of these properties can coexist in a single relation.	**3**	(CO4, L4)
2) a)	Determine whether the relation R = {(a, b) | a ≡ b (mod 5)} on the set of integers is reflexive, symmetric, antisymmetric, and transitive. Justify your answer for each property.	**3**	(CO4, L3)
	OR		
b)	Determine whether the relation R = {(a, b) | a ≤ b} on the set of natural numbers N is reflexive, symmetric, antisymmetric, and transitive. Prove your answers.	**3**	(CO4, L3)
3)	Determine whether the relation R = {(a, b) | a mod 4 = b mod 4} is an equivalence relation. If it is, find one equivalence class for the integer 13 under this relation.	**3**	(CO4, L3)
4)	Construct the matrix representation of the symmetric relation R = {(1, 2), (2, 1), (3, 4), (4, 3)} on the set A = {1, 2, 3, 4}. Also, determine if this matrix is symmetric.	**3**	(CO3, L3)
5) a)	Discuss why the set of prime numbers is an infinite countable set.	**3**	(CO5, L2)
	OR		
b)	Describe the process of finding the modular inverse of an integer a under modulo m, and explain its significance in number theory.	**3**	(CO5, L2)
6)	Determine whether the relation R = {(a, b) | a ≡ b (mod 3)} is reflexive, symmetric, antisymmetric, or transitive. Construct the matrix representation of this relation for the set {0, 1, 2} and explain its properties.	**3**	(CO4, L3)

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7) a)	Explain why the set of even natural numbers is a countable set and provide an example of a bijection between this set and the set of natural numbers.	**3**	(CO1, L2)
	OR		
b)	Explain with an example how a symmetric relation differs from an asymmetric relation.	**3**	(CO1, L2)
8)	Analyze and differentiate between a symmetric matrix and a skew-symmetric matrix by providing an example of each and explaining their key properties.	**3**	(CO3, L4)
	Section B		**8*7=56**
9)	Discuss and provide examples to explain the concept of countable sets, surjective functions, and their relationship in proving that certain sets are countable. Specifically, (i) define what it means for a set to be countable. (ii) Explain with an example how a surjective function can be used to prove a set is countable. (iii) Prove that the set of all rational numbers between 0 and 1 inclusive is countable by finding an appropriate surjection.	**8**	(CO2, L2)
10) a)	Analyze and compare the properties of modular arithmetic and set theory in solving problems related to integers. Specifically, consider a finite set S = {3, 5, 7, 11} and a relation R defined on S where (a, b) R if and only if a ≡ b (mod 4). Determine the reflexive, symmetric, and transitive properties of this relation R. Then, using your findings, explain how these properties relate to common operations in modular arithmetic.	**8**	(CO1, L4)
	OR		
b)	Examine and differentiate between equivalence relations and partial orderings using examples. For each concept, describe the properties that define it, provide an example of a set along with a relation R satisfying these properties, and explain how this relation could be represented graphically.	**8**	(CO1, L4)
11)	Consider the matrix representation of two relations, R1 and R2, defined over a set A = {a, b, c, d} as follows: R1 = {(a, a), (b, b), (c, c), (d, d), (a, b)} R2 = {(a, c), (b, d), (c, a), (d, b)}. (i) Determine the matrix representations of R1 and R2. (ii) Compute the union, intersection, and composition of R1 R2, R1 ∩ R2, and R1 R2. Represent each resulting relation using both its set notation and matrix form. (iii) Prove that the relation R1 R2 is reflexive if and only if at least one of R1 or R2 is reflexive.	**8**	(CO3, L3)
12) a)	Discuss and provide examples to explain the connection between equivalence relations and partial orderings, particularly focusing on their properties and applications in modular arithmetic. Consider a set S = {0, 1, 2, ..., 11} under modulo 12 operations. (i) Define an equivalence relation R that partitions S into three distinct equivalence classes based on congruence modulo 3. (ii) Define a partial ordering ≤ that represents divisibility for the elements in S. (iii) Describe how these relations can be represented using matrices and interpret their meaning.	**8**	(CO4, L2)
	OR		
b)	Consider two sets, A = {1, 2, 3} and B = {a, b, c}. Define a relation R from set A to set B as follows: R = {(1, a), (1, b), (2, b), (2, c), (3, c)}. Discuss whether the relation R is symmetric. If not, provide an example of a pair that violates symmetry. Then, illustrate how you would represent this relation using a matrix and a directed graph.	**8**	(CO4, L2)

13)	Analyze and differentiate between a symmetric matrix and an antisymmetric matrix by providing examples for each type. Further, examine how these types of matrices can represent symmetric and antisymmetric relations on a set A = {1, 2, 3}. Provide relation matrices to support your analysis.	**8**	(CO3, L4)
14) a)	Analyze and differentiate between an equivalence relation and a partial ordering on a given set A = {1, 2, 3, 4} by considering the following relations R and S defined over A. Determine if each is reflexive, symmetric, antisymmetric, transitive, and explain whether they represent an equivalence relation or a partial order. (i) Relation R: R = {(1,1), (2,2), (3,3), (4,4), (1,2), (2,1), (3,4), (4,3)} (ii) Relation S: S = {(1,1), (2,2), (3,3), (4,4), (1,2), (2,1), (3,4), (4,3), (1,3), (3,1), (2,4), (4,2)}	**8**	(CO4, L4)
	OR		
b)	Examine and differentiate between reflexive, symmetric, antisymmetric, and transitive relations by analyzing the properties of the relation R = {(a, b) | a - b is even} over the set of integers Z. Provide examples to support your analysis.	**8**	(CO4, L4)
15)	Explain with examples how an injection differs from a surjection and provide conditions for each in terms of domain and codomain mappings. Also, discuss the implications of a bijective function on the sets involved.	**8**	(CO2, L2)
	Section C		**10*2=20**
16)	Determine and construct an example to show that a function can be injective but not surjective, and vice versa. Then, for each case, find an explicit example of such functions defined on finite sets A and B, where |A| = 3 and |B| = 4. Explain why your examples satisfy the conditions.	**10**	(CO2, L3)
17) a)	Given two integers a = 72 and b = 144, demonstrate a working knowledge of set theory by constructing a set A containing all common divisors of a and b. Using this set, determine if the elements satisfy the properties of a symmetric relation with respect to the divisibility operation (denoted as |). Finally, calculate the greatest common divisor (gcd) of a and b using the Euclidean algorithm.	**10**	(CO1, L3)
	OR		
b)	Determine whether the set of all rational numbers between 0 and 1, inclusive, is countable or uncountable. If you conclude it is countable, provide an example of a bijection mapping the natural numbers to this set. If not, demonstrate why it cannot be put into a one-to-one correspondence with the natural numbers.	**10**	(CO1, L3)

Revised Bloom's Taxonomy (RBT) Levels

L1 - Remember
L2 - Understand
L3 - Apply
L4 - Analyse
L5 - Evaluate
L6 - Create